

GAUTAM RAO

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EDUCATION

University of Texas at Austin

Master of Science, Computational Engineering

May 2027

Austin, TX

University of Texas at Austin

Bachelor of Science, Computational Engineering

May 2026

Austin, TX

TECHNICAL EXPERIENCE

84.51

May 2026 - Aug 2026

Data Science Intern

- Built a reusable Optuna-based hyperparameter optimization framework to tune Kroger's search ranking model and translate experiment results into improved trading-style decision inputs.
- Improved ranking quality by 1% and conversion by 0.5%, generating an estimated \$60M in incremental revenue and 7.8M additional items added to cart by users annually.
- Executed 100 trials per experiment, analyzed model performance across configurations, and reduced tuning time by 88%, enabling faster experimentation and model iteration.
- Evaluated training-data sampling strategies, feature weighting, and model configuration changes through mathematical modeling and quantitative analysis, identifying configurations that boosted ranking quality.
- Leveraged Databricks and Azure for machine learning experimentation, data processing, model training, and workflow execution, streamlining the end-to-end pipeline and reducing processing latency.
- Identified multiple GPU cluster limitations and coordinated with Databricks team and Microsoft to secure more efficient compute resources, reducing model tuning runtime and improving performance.

Collins Aerospace

Jan 2024 - Aug 2025

Data Science Analyst

- Implemented predictive maintenance and anomaly-detection models using operational and equipment data, reducing troubleshooting time by 35% and supporting \$750K in annual savings.
- Engineered SQL and ETL pipelines across Snowflake, SQL Server, SAP, and Alteryx to clean, integrate, and transform aerospace operations data for downstream statistical modeling and analysis.
- Designed and optimized relational databases using advanced SQL, improving query performance by 40%, data accessibility, and reporting efficiency across multiple operational workflows.
- Developed image classification workflows using Spark and PyTorch/PyTorch Lightning to detect defective parts, reducing quality-control review time by 40%.
- Automated Tableau and Power BI dashboards for recurring operational reporting, saving 25+ hours per week in manual analysis and increasing reporting refresh frequency.
- Partnered with engineering, avionics, and operations teams to translate model outputs into actionable insights for equipment monitoring, root-cause analysis, risk assessment, and process improvement.

PROJECTS

Amazon Recommendation System

Aug 2023

MIT

- Built a Python-based product recommendation system using collaborative filtering and matrix factorization to model user-product preferences from explicit rating data.
- Implemented SVD-based recommendation models and evaluated predictive performance using F1 score, precision, and recall to compare model effectiveness.
- Achieved an F1 score of 0.852, precision of 0.837, and recall of 0.868 using Scikit-learn, TensorFlow, and PyTorch.

SKILLS

- **Languages & Data:** Python, SQL, C++, R, Rust, SQL Server, Spark, Snowflake, Databricks, MATLAB
- **Machine Learning:** Scikit-learn, Pandas, NumPy, TensorFlow, PyTorch, Predictive Modeling, Recommendation Systems, Ranking Models, Feature Engineering, Statistical Modeling, LLM, XGBoost, Optuna, Mathematical Modeling
- **Tools & Cloud:** Alteryx, Git, Tableau, Power BI, JupyterLab, Docker, Kubernetes, Azure
- **Certifications:** MIT IDSS Data Science and Machine Learning Certificate, IBM Data Science Professional Certificate, Google Advanced Data Analytics Professional Certificate
- **Patents:** US Patent Holder for ALS Device: US20230324680A1
- **Financial Markets:** Financial Markets